

SQL Learning Journal #17- INNER (Self) JOIN

1. INNER (Self) JOIN function

There is no special `SELF JOIN` command in SQL. A self join is created by using a regular `INNER JOIN` to join a table to itself. By using different aliases for the same table, we can compare or connect information from different rows within that table.

table_1:

product_id	product	category_id
1	laptop	10
2	mouse	10
3	keyboard	10
4	coffee	15
5	tea	15
6	milk	15

SELECT

t1.product_name AS product1

t2.product_name AS product2

FROM table_1 **AS** t1

INNER JOIN table_1 **AS** t2

ON t1.category_id=t2.category_id

AND t1.product_id<t2.product_id

The following condition is important:

AND t1.product_id < t2.product_id

It prevents a product from being matched with itself and also prevents the same pair from appearing twice in reverse order.

Result_table:

product1	product2
laptop	mouse
laptop	keyboard
mouse	keyboard
coffee	tea
coffee	milk
tea	milk

The query therefore finds pairs of different products that belong to the same category.

What I Learned Today

At first, it was difficult to understand how the same table could be used as two tables in a query. Today I learned that a SELF JOIN uses the same table with different aliases, allowing us to compare or connect different rows within that table. I also learned how an additional condition can prevent a row from being matched with itself and avoid duplicate pairs.